



DEALER BEST PRACTICE SERIES

CRC Investment Analysis

Process – Measuring

Return on Investment

Application Maintenance Site Management **Component Rebuild** Safety MARC Management

CRC Investment Analysis0

Process – Measuring Return on Investment0

1.0 Introduction1

2.0 Best Practice Description.....1

3.0 Implementation Steps2

4.0 Benefits.....3

5.0 Resources Required3

6.0 Supporting Attachments / References3

7.0 Related Best Practices3

8.0 Acknowledgements.....3

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

Most dealerships require a formal return on investment analysis be performed to measure the expected return on any proposed significant investment. This process is necessary to assure that company funds are used wisely and create a satisfactory return on the investment.

It is important, especially in the case of investments made in the Component Rebuild Center (CRC) or other service area, to be sure that the full impact of the investment is properly measured and analyzed. Important benefits of service shop investments are easily overlooked and thus decisions are sometimes made on incomplete data.

This Best Practice compiles a list of potential benefits to consider when evaluating a CRC investment. The ideas have been combined from the practices of several dealerships.

2.0 Best Practice Description

CRC Investments are typically made to achieve a performance result leading to a profit improvement. Here are the most common reasons for shop investments:

- **Improved Labor Efficiency.** This is a process change to reduce the labor required to perform a service. It is important to recognize that a labor efficiency improvement also has impact on profit margins, shop capacity, and work order turnaround time (velocity). An example of this type of investment is tooling improvements, which reduce the time a mechanic takes to perform a task.
- **Improved Rebuild Quality.** A rebuild quality improvement lowers warranty costs and creates a cost-avoidance benefit. If a dealer has MARC contracts, a quality improvement also increases rebuild reliability and component life. Component life improvement reduces cost per hour and can mean substantial financial benefit. A good example is testing/inspection equipment or contamination control equipment.
- **Cost Reduction.** This is an improved process or technology that lowers shop operating costs.
- **Revenue Creation.** Investments made to increase sales.
- **Add Capability.** An investment made to provide a service or shop support capability not currently available. This can create new revenue or lower cost. A good example is investing in spray weld or machine shop technology.
- **Increase Capacity.** Such investments improve processes and efficiency, resulting in lower work order turnaround time. Examples include adding additional facility space or technicians.
- **Support Marketing.** Service capability is often important in supporting the marketing efforts of new machine sales, parts sales, MARC contracts, etc.
- **Enhance Image.** An investment to improve both internal and external image. Internal image is an environment that serves to motivate technicians/ staff towards higher performance at many levels, while external image projects quality and professionalism to the customer.

Many of the above improvements (investments) result in a direct, or indirect, financial benefit.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
CRC Investment Analysis Process - Measuring Return on Investment	DATE 2/19/2015	CHG NO 01	NUMBER 0107-4.5-1054

The typical investment analysis is a basic cash flow forecast. It may or may not capture the full impact of an investment on the bottom line. It is important that any investment under consideration be fully explored to capture both the obvious and the “hidden” value.

3.0 Implementation Steps

The following is a collection of investment analysis process “tips” successfully used by a number of dealers.

- Flat rates are vital.** It is critical to apply flat rated standard segments, at least for the labor portion, to all work orders. For example, a certain repair may require 10 hours labor to perform. Tooling improvements may allow the job to be performed in eight hours labor in the future. If the charge to the customer is reduced to eight hours, you actually end up selling fewer hours and will never pay for the tooling. If you charge the 10-hour flat rate and do the work in eight hours, you have significantly increased the profit margin on the job.
Note: You may also be able to build a financial case for charging only the 8 hours if this lower price to your customers creates more business and you can sell the 2 hours that you saved on another job.
- CRC or shop investments may be assessed at the shop and/or the enterprise impact level.** Labor revenue alone may not be sufficient to justify a significant investment, but there may be influence on parts sales as well as machine sales. *A poor investment from the shop perspective may be a good investment from the company perspective.*
- Be very thorough with capturing the benefits of efficiency improvements.** Labor efficiency improvements have a compounded impact on profits. In the above-mentioned example of a 10-hour job, you invest in tooling to do the job in 8 hours and charge a flat rate equal to 10 hours. The 2 hours that you saved becomes pure profit at your full selling rate and substantially changes your profit margin for that job. Be sure that your investment analysis spreadsheet reflects the higher margin. Now, the 2 hours saved effectively increase your shop capacity and can be sold on other jobs. This additional revenue should also be captured in your analysis.
- Measure cost-avoidance benefits.** Reduced costs and cost avoidance goes directly to your bottom line as increased profit. Cost avoidances are hard to capture, but are very real. Improved quality lowers warranty costs and increases average component life. Lower warranty cost improves your profits, and if you have CSAs, better average component life reduces your contract expenses.
- Utilization of assets benefit.** Better use of assets can be a major benefit. If you improve efficiencies and processes to reduce component turnaround time, you have added capacity to your shop using the same resources. This enables your shop to handle more workload. One dealer cancelled plans for a \$1.5 million dollar shop expansion when they were able to improve shop turnaround time 25%. The capacity improvement made the investment unnecessary at present and forecast workloads. Another dealer reduced their major component exchange inventory by several million dollars due to improved component rebuild turnaround time in the shop. The exchange components were to support a fleet of 793's. Better utilization of assets represents a benefit that needs to be part of your analysis.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

CRC Investment Analysis Process - Measuring Return on Investment	DATE 2/19/2015	CHG NO 01	NUMBER 0107-4.5-1054
---	-------------------	-----------------	-------------------------

- **Future machine population.** Most investment analysis spreadsheets look at a 10-year return on investment. Looking out ten years, your analysis should reflect shop business volume considering present and forecast machine populations.

Improvement investments should always be carefully considered. Cash flow forecasts typically have assumptions built into the calculations such as margins, volume of sales, etc. When investment impacts efficiency, capacity, turnaround time, quality, cost reductions, or any other etc, it is important to ensure that the financial forecast, used to calculate return on investment, reflects the improvement created by the investment.

4.0 Benefits

- Investment analysis is performed capturing all costs and benefits.
- Shop as well as enterprise impact is considered.

5.0 Resources Required

N/A

6.0 Supporting Attachments / References

N/A

7.0 Related Best Practices

Improving CRC Rebuild Turnaround Time 0906-4.5-1013

8.0 Acknowledgements

This Best Practice was written by:

Dale Bream
6 Sigma Black Belt
Global Mining Product Support
brehm_dale_e@cat.com
309 -675-6325

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

CRC Investment Analysis Process - Measuring Return on Investment	DATE 2/19/2015	CHG NO 01	NUMBER 0107-4.5-1054
---	-------------------	-----------------	-------------------------